Cambridge Assessment International Education

# Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*9518465764*

**BIOLOGY** **9700/34**

Paper 3 Advanced Practical Skills 2 **October/November 2020**

**2 hours**

You must answer on the question paper.

You will need: The materials and apparatus listed in the confidential instructions

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

**INFORMATION**

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [ ].

| For Examiner's Use | |
|---|---|
| 1 | |
| 2 | |
| Total | |

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[Turn over

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Before you proceed, read carefully through the **whole** of Question 1 and Question 2.

Plan the use of the **two hours** to make sure that you finish the whole of Question 1 and Question 2.

**1** Amylase is an enzyme which hydrolyses starch into reducing sugars, as shown in Fig. 1.1.

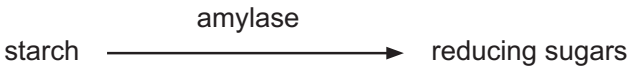



**Fig. 1.1**

The progress of this reaction can be followed by measuring the concentration of starch remaining using iodine solution.

You will need to:

- prepare a serial dilution of a 1.0% starch solution, **S**
- estimate the concentration of starch remaining after hydrolysis with amylase at two different pH values.

You are provided with the materials shown in Table 1.1.

**Table 1.1**

| **labelled** | **contents** | **hazard** | **volume / $cm^3$** |
|---|---|---|---|
| **S** | 1.0% starch solution | none | 50 |
| **iodine** | 0.01 mol $dm^{-3}$ iodine solution | none | 20 |
| **W** | distilled water | none | 150 |
| **X** | a sample of 1.0% starch solution which has been incubated with amylase and pH2 buffer | harmful irritant | 20 |
| **Y** | a sample of 1.0% starch solution which has been incubated with amylase and pH7 buffer | harmful irritant | 20 |

If any solution comes into contact with your skin, wash off immediately under cold water.

It is recommended that you wear suitable eye protection.

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**[Turn over**

**(a)** You need to carry out a **serial** dilution of the 1.0% starch solution, **S**, which reduces the concentration of the starch solution by a factor of 10 between each successive dilution.

Fig. 1.2 shows you the first two beakers you will use to make your serial dilution.

**(i)** Complete Fig. 1.2 by drawing as many extra beakers as you need for your serial dilution.

For each beaker:

- state, under the beaker, the volume and concentration of starch solution available for use in the investigation
- use one arrow with a label, above the beaker, to show the volume and concentration of starch solution added to prepare the concentration
- use another arrow with a label, above the beaker, to show the volume of **W** added to prepare the concentration.

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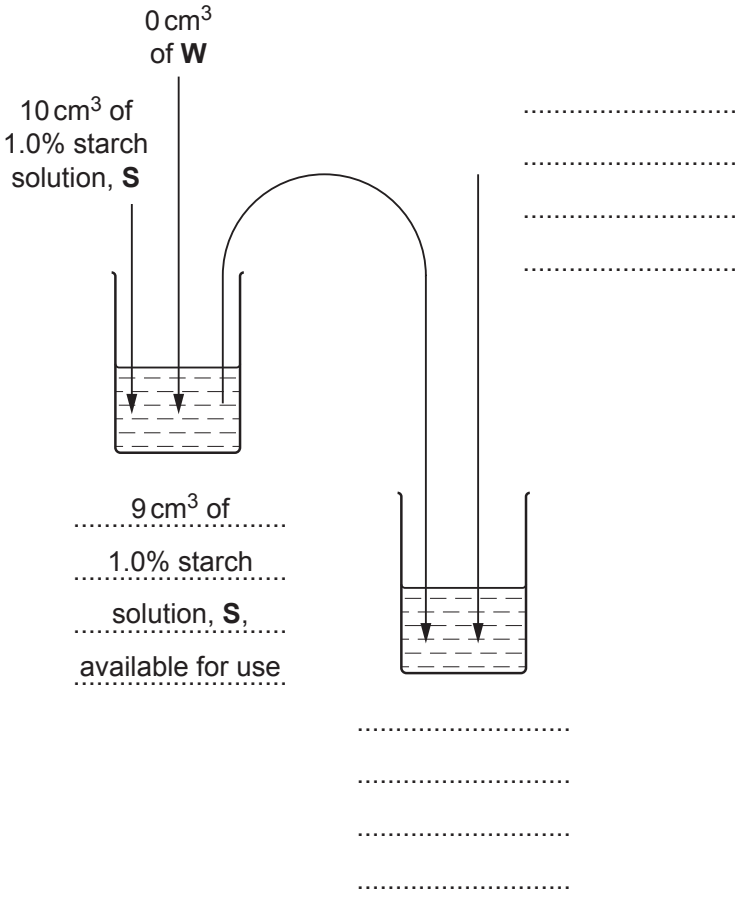

0 cm3 of W
10 cm3 of 1.0% starch solution, S
9 cm3 of 1.0% starch solution, S, available for use

**Fig. 1.2**

[3]

Carry out step 1 to step 6.

1. Prepare the concentrations of starch solution as decided in **(a)(i)** and shown in Fig. 1.2.

   Use a glass rod to mix the starch solutions and water.

2. Label test-tubes with the concentrations of starch solution prepared in step 1.

3. Put 5 $cm^3$ of each concentration of starch solution into an appropriately labelled test-tube.

4. Put 3 drops of **iodine** into each of the test-tubes. Shake gently to mix.

5. After step 4, observe the colour of the liquid in each test-tube. You may see the same colour in more than one test-tube.

It may help to hold a piece of white card behind the test-tube to see the colour more clearly.

Fig. 1.3 shows the key you need to use to record your results.

**Key**

| **colour** | **symbol** |
|---|---|
| blue/black or purple | +++++ |
| blue | ++++ |
| dark brown | +++ |
| brown | ++ |
| yellow/orange | + |

**Fig. 1.3**

6. Record your results in **(a)(ii)** using the symbols shown in the key in Fig. 1.3.

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**(ii)** Record your results in an appropriate table.

[5]

Carry out step 7 to step 10.

7. Label a test-tube **X**.

8. Put 5 $cm^3$ of **X** into this test-tube.

9. Put 3 drops of **iodine** into this test-tube. Shake gently to mix.

10. Observe the colour of the liquid and record the result in **(a)(iii)** using the symbols shown in Fig. 1.3.

11. Repeat step 7 to step 10, using **Y** instead of **X**.

**(iii)** Record your result for **X** and **Y** using the symbols shown in the key in Fig. 1.3.

result for **X** .....

result for **Y** .....

[1]

**(iv)** Using your results in **(a)(ii)** and **(a)(iii)**, estimate the concentration of starch in **X** and **Y**.

**X** = .....

**Y** = .....

[1]

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**[Turn over**

**(v)** A student investigated the hydrolysis of starch by amylase.

The student decided to use a semi-quantitative method to measure the products.

These products are reducing sugars.

Suggest how the student would use a semi-quantitative method to measure the concentration of reducing sugars produced.

.....

.....

.....

.....

.....

.....

..... [3]

**(b)** One of the reducing sugars produced by the hydrolysis of starch is maltose.

Maltose is a disaccharide that can be hydrolysed to glucose by the enzyme maltase.

A student investigated the effect of maltose concentration on the initial rate of reaction of maltase.

The results are shown in Table 1.2.

**Table 1.2**

| concentration of maltose / mol dm$^{-3}$ | initial rate of reaction of maltase / arbitrary units |
|---|---|
| 0.00 | 0 |
| 0.60 | 90 |
| 1.30 | 175 |
| 1.85 | 225 |
| 2.45 | 255 |
| 3.55 | 260 |
| 3.95 | 260 |

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**(i)** Plot a graph of the data in Table 1.2 on the grid in Fig. 1.4.

Use a sharp pencil for drawing graphs.

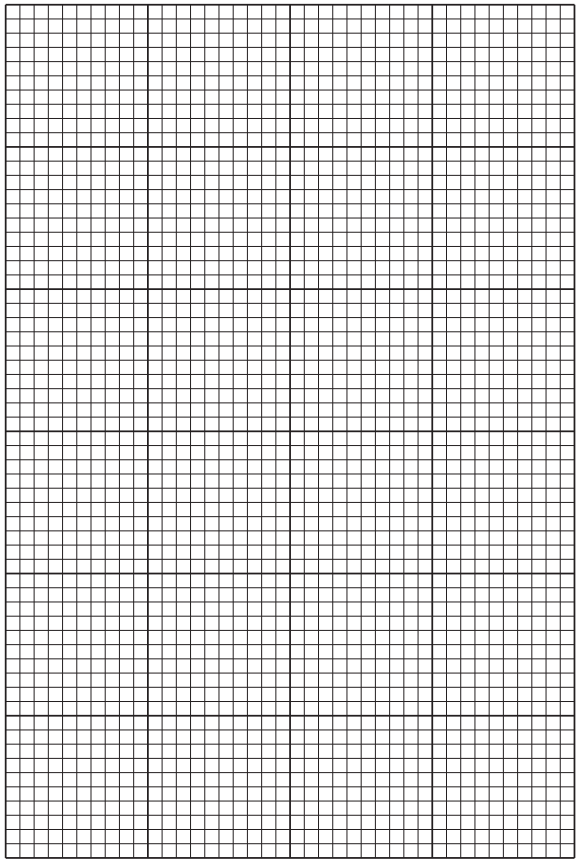


**Fig. 1.4**

[4]

**(ii)** Use the graph in Fig. 1.4 to estimate the Michaelis–Menten constant, $K_m$.

Show your working on the graph and in the space below.

$K_m$ = ..... mol $dm^{-3}$ [2]

**(iii)** A student investigated a different enzyme, **Z**, and found the $K_m$ value to be 0.50 mol $dm^{-3}$.

State which enzyme, **Z** or maltase, has a **lower affinity** for its substrate. Give a reason for your answer.

enzyme .....

reason .....

.....

[1]

**(iv)** Using the graph in Fig. 1.4, explain the change in the rate of reaction between 0.00 mol $dm^{-3}$ of maltose and 3.95 mol $dm^{-3}$ of maltose.

.....

.....

.....

.....

.....

..... [2]

[Total: 22]

**2** **L1** is a slide of a stained transverse section through a plant stem.

You are not expected to be familiar with this specimen.

**(a)** Select a field of view so that you can observe the different tissues as shown by the shaded area in Fig. 2.1.

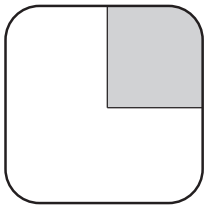


**Fig. 2.1**

Use a sharp pencil for drawing.

You are expected to draw the correct shape and proportions of the different tissues.

**(i)** Draw a large plan diagram of the region of the stem on **L1** shown by the shaded area in Fig. 2.1.

Use **one** ruled label line and label to identify the xylem.

[5]

**(ii)** Observe the cells in the central tissue (pith) of the stem section on **L1**.

Select **four** adjacent, touching cells which show observable features of the central tissue.

Each cell must touch at least two other cells.

- Make a large drawing of this group of **four** touching cells.
- Use **one** ruled label line and label to identify the cell wall of **one** cell.

[5]

**(b)** Fig. 2.2 is a photomicrograph of a stained transverse section through a stem of a different type of plant.

You are not expected to be familiar with this specimen.

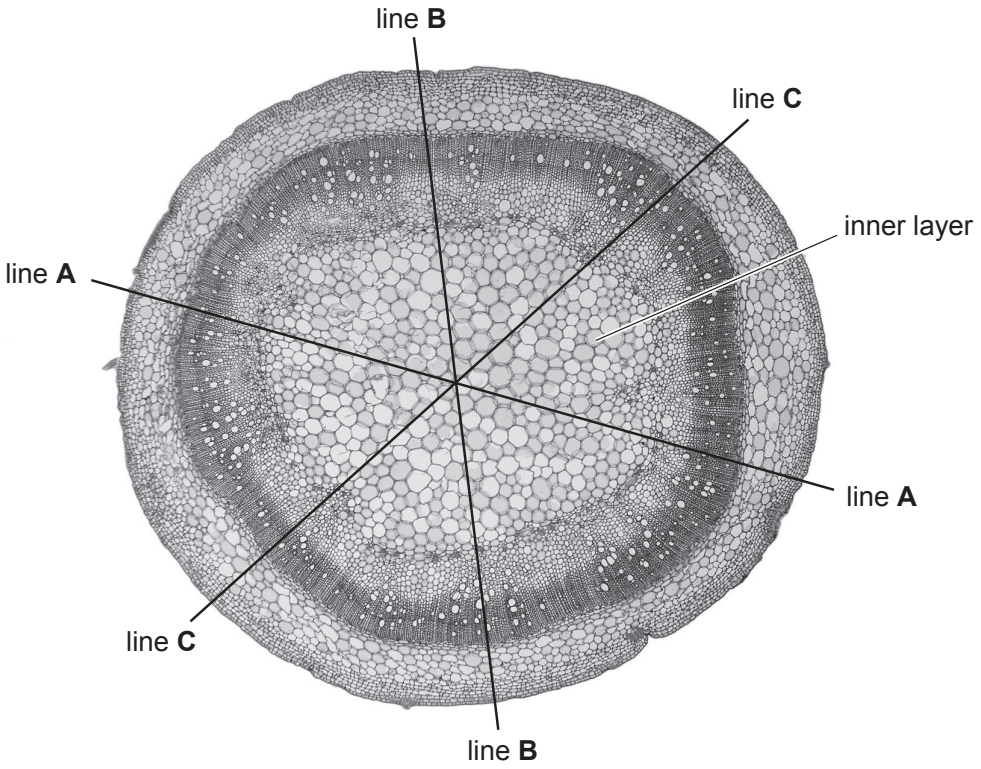



**Fig. 2.2**

**(i)** The magnification of the stem section in Fig. 2.2 is ×32.
Use line **A**, line **B** and line **C** to determine:

- the mean actual diameter of the whole stem section
- the mean actual diameter of the inner layer.

Show all the steps in your working.

mean actual diameter of whole stem section .....

mean actual diameter of inner layer .....

[4]

**(ii)** Use your answers to **(b)(i)** to determine the simplest whole number ratio of the mean actual diameter of the whole stem section to the mean actual diameter of the inner layer.

simplest whole number ratio = ..... [1]

**(c)** Fig. 2.3 is a photomicrograph of the same stem section that is in Fig. 2.2.

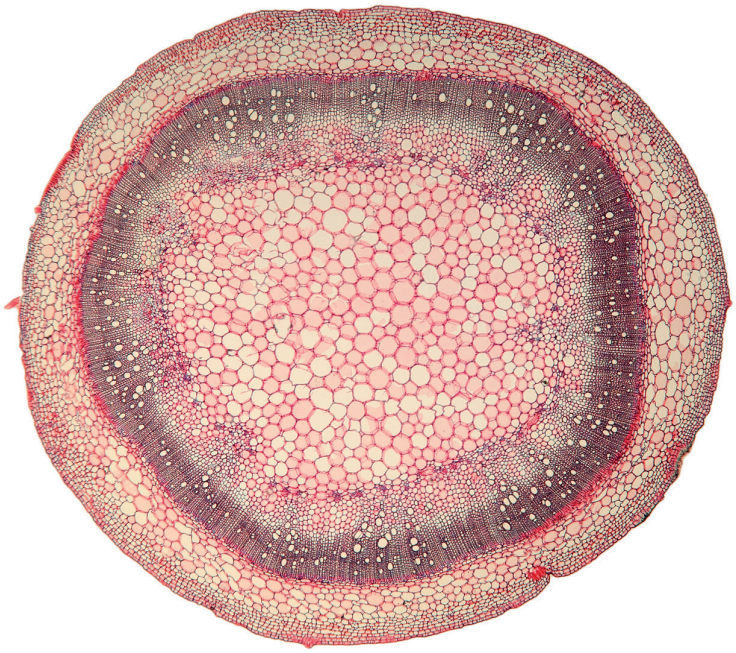


**Fig. 2.3**

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Identify the observable differences between the stem section in Fig. 2.3 and the stem section on **L1**.

Record the observable differences in Table 2.1.

**Table 2.1**

| feature | Fig. 2.3 | L1 |
|---|---|---|
| | | |

[3]

[Total: 18]

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